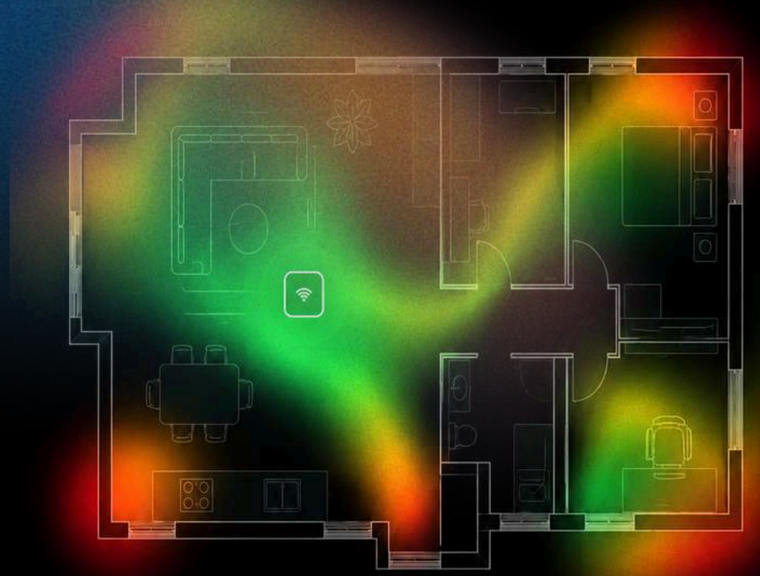
WIFI DEADZONE MAPPER

Floor-Plan-Based Wi-Fi Analysis & Portable ESP8266 Repeater



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PROBLEM STATEMENT

Physical obstructions create unpredictable WiFi dead zones that standard signal bars fail to map across a building. As a result, users buy or move extenders blindly without knowing where coverage actually fails or where hardware will be effective. Solving this requires an integrated approach to accurately map spatial signal strength and dynamically extend coverage to identified dead zones.

Why Wi-Fi Dead Zones Matter

- Wi-Fi coverage affects everyday activities that depend on a stable wireless connection. A network may work properly in one room while becoming less reliable in another.

- Why mapping helps:

  Instead of guessing where Wi-Fi is weak, users can compare measurements from different locations and use the floor plan to understand the spatial pattern.

  The project focuses on identifying and visualizing the problem before attempting to improve coverage

PROJECT IDEA AND OBJECTIVES

- THE PROJECT COMBINES TWO COMPLEMENTARY COMPONENTS:

1. WI-FI DEADZONE MAPPER ANDROID APPLICATION

The application provides Wi-Fi network information and a time-based signal graph. The planned floor-plan workflow allows the user to upload a floor-plan image, collect measurements at different positions, and represent the results as a heatmap.

2. ESP8266-BASED WI-FI REPEATER

A NodeMCU ESP8266 board is configured with repeater firmware. The prototype is powered through a battery and a voltage-boosting power module, allowing a portable hardware setup.

- The project focuses on monitoring nearby Wi-Fi signals, mapping coverage through floor plans and generating heatmaps, and identifying optimal repeater locations. It also involves developing an ESP8266 repeater prototype and evaluating coverage improvements before and after deployment.

TARGET AUDIENCE

HOME USERS

Identify rooms with consistently weak Wi-Fi coverage, assess the impact of router placement, and determine optimal repeater or extender locations.

STUDENTS & HOSTEL RESIDENTS

Investigate weak Wi-Fi coverage in study areas, bedrooms, and shared spaces while gaining hands-on experience with wireless networking.

HOSPITAL HEALTHCARE FACILITIES

Ensuring uninterrupted connectivity for critical medical IoT devices (e.g., patient monitors, infusion pumps).

WAREHOUSES & FACTORIES

Wi-Fi heatmaps identify dead zones affecting inventory scanners, machinery, and IoT sensors in factories and warehouses. Wi-Fi extenders improve connectivity for reliable asset tracking and real-time monitoring.

WHAT IS A WI-FI REPEATER?

A Wi-Fi repeater is a device that receives a wireless network signal and retransmits it so that connectivity can reach areas beyond the original router's useful coverage.

BASIC WORKING PRINCIPLE

1. The main router provides the original Wi-Fi network.
2. The repeater is placed where it can still receive a usable signal from the router.
3. The repeater connects to the original network according to its configured operating mode.
4. It retransmits the connection, creating an additional area of Wi-Fi coverage.
5. Devices in the extended area connect through the repeater when their connection and configuration permit it.

In this project, the ESP8266 board is configured using repeater firmware rather than being programmed as a custom repeater from scratch.

OVERALL SYSTEM DESIGN

The project consists of a software component and a hardware component, connected by a common objective: improving the understanding and reach of Wi-Fi coverage.



ANDROID APPLICATION

- Scans and displays available access points.
- Displays a signal-strength graph over time for a selected access point.
- Provides a floor-plan heatmap workflow for spatial coverage analysis.
- Helps the user identify locations requiring further investigation.

ESP8266 REPEATER

- Uses a NodeMCU ESP8266 development board.
- Runs prebuilt repeater firmware.
- Is configured through its device-hosted web interface.
- Uses a battery-powered arrangement with a booster/power module.

USER INTERFACE

The application is organized around three main views, each serving a different stage of Wi-Fi analysis

1. ACCESS POINT LIST

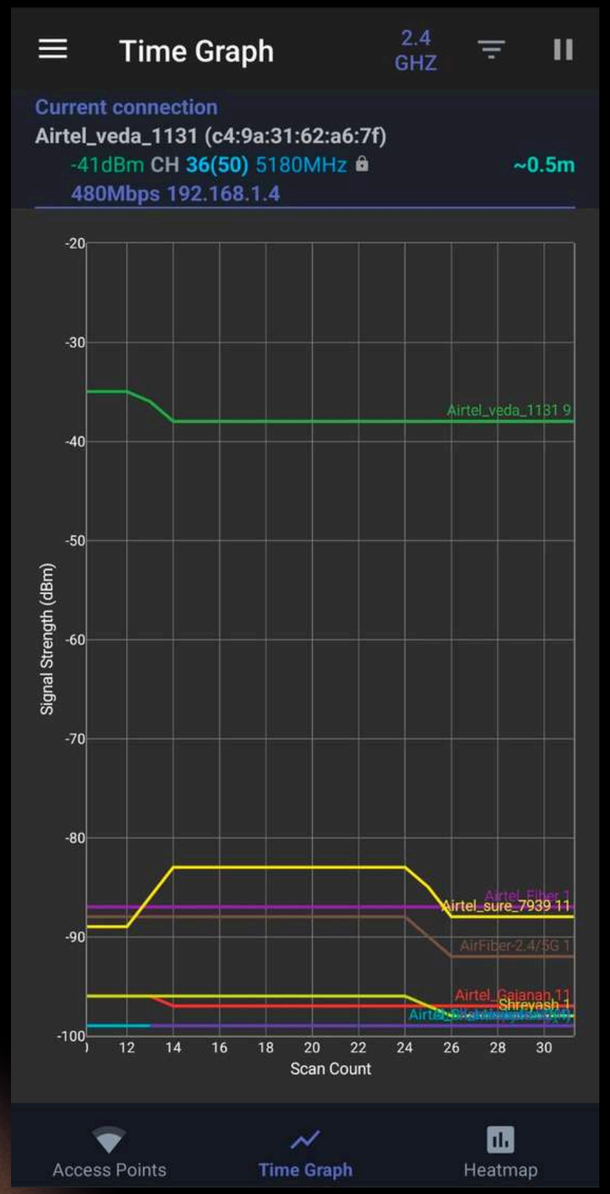
1. Displays available Wi-Fi access points detected by the application.
2. Allows the user to inspect network names and the available network details.
3. Provides the starting point for selecting a network to observe.
4. Helps distinguish between the presence of a network and its signal quality at the current location.



USER INTERFACE

2. TIME GRAPH

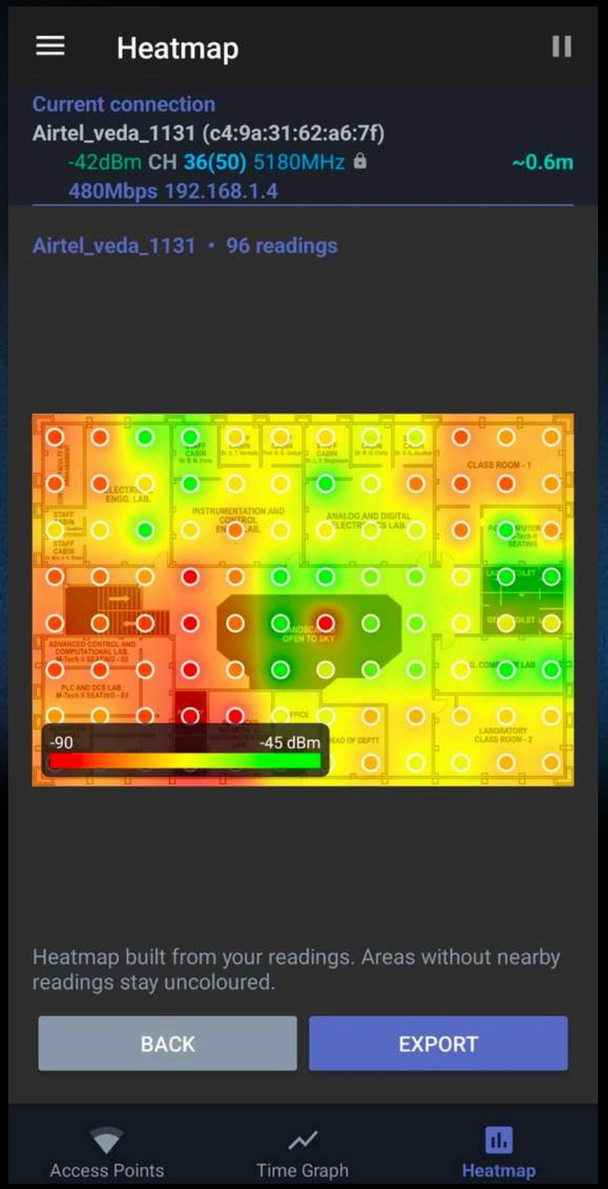
1. Displays the selected access point's signal strength over time.
2. Helps reveal whether the signal remains relatively stable or fluctuates while the user moves or changes position.
3. Supports observation of short-term changes during a measurement session.



USER INTERFACE

3. HEATMAP

- Provides the intended floor-plan-based coverage visualization workflow.
- The user uploads a floor-plan image.
- Measurements are associated with locations on the floor plan.
- The collected data is represented visually to distinguish stronger and weaker coverage areas.



FROM HEATMAP TO REPEATER PLACEMENT :

STEP 1: IDENTIFY THE WEAK REGION

STEP 2: EXAMINE THE SURROUNDING AREA

STEP 3: SELECT A CANDIDATE LOCATION

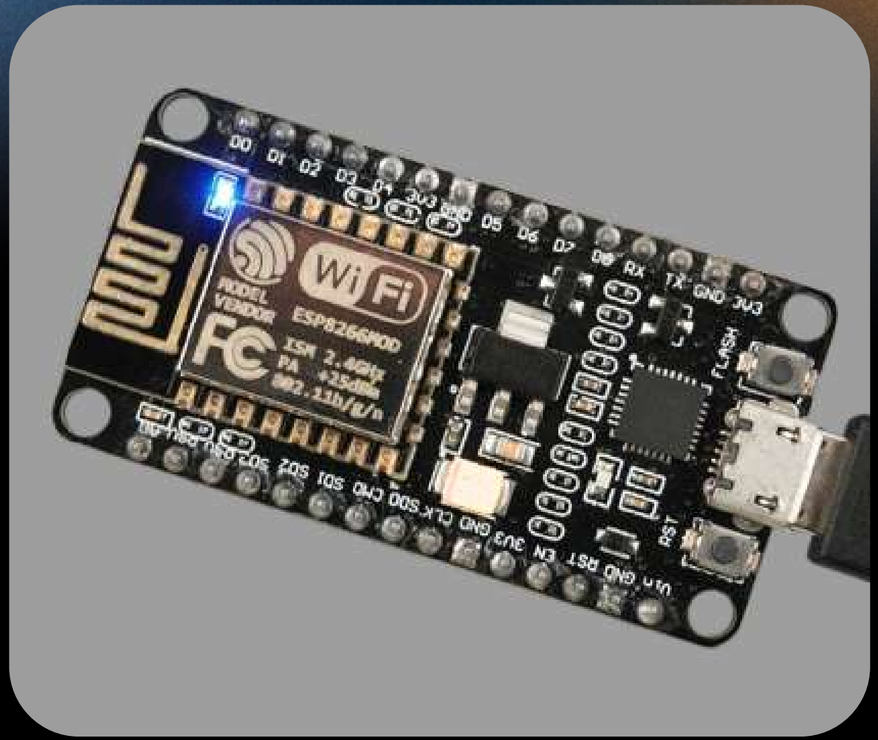
STEP 4: CONFIGURE THE REPEATER

STEP 5: MEASURE AGAIN

HARDWARE COMPONENTS

The hardware prototype uses a NodeMCU ESP8266 development board, a single 18650 battery cell, a battery holder, and a power module to create a portable repeater arrangement.

1. **ESP8266 NodeMCU :**
    - Acts as the main controller
    - Runs the repeater firmware
    - Provides wireless connectivity



HARDWARE COMPONENTS

2. POWER BOOSTER MODULE



- Provides the power-conversion arrangement used in the prototype
- To boost a low DC voltage to a higher voltage

3. 18650 LI-ION BATTERY CELL



- Portable energy source
- Provides 3.7v voltage

HARDWARE COMPONENTS

4. Micro-USB cable



- Connects the power module to the ESP8266 board

5. 18650 battery holder



- Securely hold the 18650 Li-ion battery
- Connects it electrically to the power circuit

PROTOTYPE DEVELOPMENT

Hardware Prototype — Setup Completed

- Configured NodeMCU ESP8266 with repeater firmware.
- Set SSID and network settings via web interface.
- Built a portable power supply using an 18650 battery and booster module.

Application Prototype — Current Features

- Wi-Fi access-point scanning and listing.
- Signal-strength monitoring over time using graphs.
- Floor-plan heatmap for spatial analysis (status depends on the demonstrated build).

Current system boundary

The application analyses Wi-Fi observations, while the repeater is configured separately. The current prototype does not establish automatic communication between the heatmap and the repeater.

TESTING PLAN

APPLICATION TESTING

- Launch & Scanning: Verify app startup and detected Wi-Fi networks.
- Signal Graph: Check signal-strength updates over time.
- Floor Plan & Heatmap: Test image upload, position labels, readings, and heatmap generation.

HARDWARE TESTING

- Hardware Testing Firmware & Configuration: Verify flashing, configuration-page access, and SSID settings.
- Repeater Operation: Confirm upstream connection and internet access.
- Battery Operation: Check stable performance on portable power.

END-TO-END TESTING

- Measure Wi-Fi strength at fixed positions before and after repeater placement.
- Compare signal readings and connectivity to assess coverage improvement.

VALIDATION METHOD

1. Application Validation
   - Verify Wi-Fi scanning and signal-strength graphs.
   - Test floor-plan upload and heatmap accuracy (if operational).
   - Compare displayed results with collected data.

2. Repeater Validation
   - Verify firmware startup and configuration-page access.
   - Test upstream Wi-Fi connection and device connectivity.
   - Repeat tests using battery power.

3. End-to-End Validation
   - Measure baseline signal strength at selected locations.
   - Identify weak-coverage areas and place the repeater strategically.
   - Remeasure at the same locations and compare signal strength and connectivity.



RESULTS & CONCLUSION

The Android application prototype supports Wi-Fi access-point listing and signal-strength monitoring through time graphs. The hardware prototype uses an ESP8266 flashed with repeater firmware, configured via 192.168.4.1, and powered by a portable 18650 battery and booster module.

CONCLUSION

The project combines Wi-Fi analysis with a practical hardware-based approach to extending coverage. The application provides a way to inspect networks and observe signal changes, while the ESP8266 repeater offers a physical method for testing coverage extension. The next step is to validate the complete workflow through repeatable measurements and a documented before-and-after comparison.

FUTURE SCOPE

COMPLETE THE FLOOR-PLAN HEATMAP WORKFLOW

INTEGRATE THE APPLICATION AND HARDWARE



IMPROVE MEASUREMENT QUALITY

IMPROVE THE PORTABLE HARDWARE

ADD BEFORE-AND-AFTER COMPARISON

IMPROVE REPEATER PLACEMENT GUIDANCE

Thank You